

DocuInsight - Executive Brief

What It Is

DocuInsight is an AI-powered Document Intelligence Platform that helps organizations assess and maintain the health of their knowledge repositories. It connects to SharePoint/OneDrive document libraries and uses advanced semantic analysis to identify duplicates, detect conflicts, and evaluate content quality.

The Problem It Solves

Enterprise document repositories grow organically over time, leading to:

- **Duplicate content** scattered across folders
 - **Conflicting information** between documents (different dates, numbers, facts)
 - **Outdated documents** that should be archived or updated
 - **Poor organization** making it hard to find authoritative information
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How It Works

1. Connect & Sync

DocuInsight connects to your SharePoint site using secure Microsoft Graph API integration. Documents are synced and their text content is extracted from PDF, Word, PowerPoint, and Excel files.

2. AI-Powered Analysis

Each document is processed through OpenAI's embedding model to create a semantic "fingerprint" - a vector representation of its meaning. These vectors are stored in a PostgreSQL database with pgvector for fast similarity searches.

3. Intelligent Detection

The platform automatically identifies:

- **Duplicates** (80%+ similarity) - Near-identical documents
- **Similar Content** (50-79% similarity) - Related documents that may need consolidation
- **Conflicts** - Contradictory information detected by GPT analysis
- **Novel Documents** - Unique content that adds new knowledge

4. Pre-Publish Checking

Before adding new documents, users can upload and check them against the existing repository. The system shows similar documents, explains overlaps, detects potential conflicts, and recommends whether to publish, review, or reject.

Key Features

Feature	Description
Health Dashboard	At-a-glance metrics showing repository quality

Document Browser	Browse and sync documents from SharePoint
Analysis View	Document-centric view of duplicates, similar content, and conflicts
Pre-Publish Check	Validate new documents before adding to repository
AI Explanations	Human-readable explanations of why documents are similar

Technology Stack

- **Frontend:** React with TypeScript, Tailwind CSS
 - **Backend:** Node.js/Express with PostgreSQL + pgvector
 - **AI:** OpenAI embeddings (text-embedding-3-small) + GPT-4o for analysis
 - **Integration:** Microsoft Graph API for SharePoint/OneDrive
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Current Deployment

- **Repository:** ValueSite SharePoint (199 documents)
 - **Detected:** 2,408 duplicate pairs, 12,599 similar pairs
 - **Performance:** Sub-second similarity searches via pgvector indexing
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DocuInsight transforms document chaos into actionable insights, helping knowledge managers maintain authoritative, conflict-free repositories.